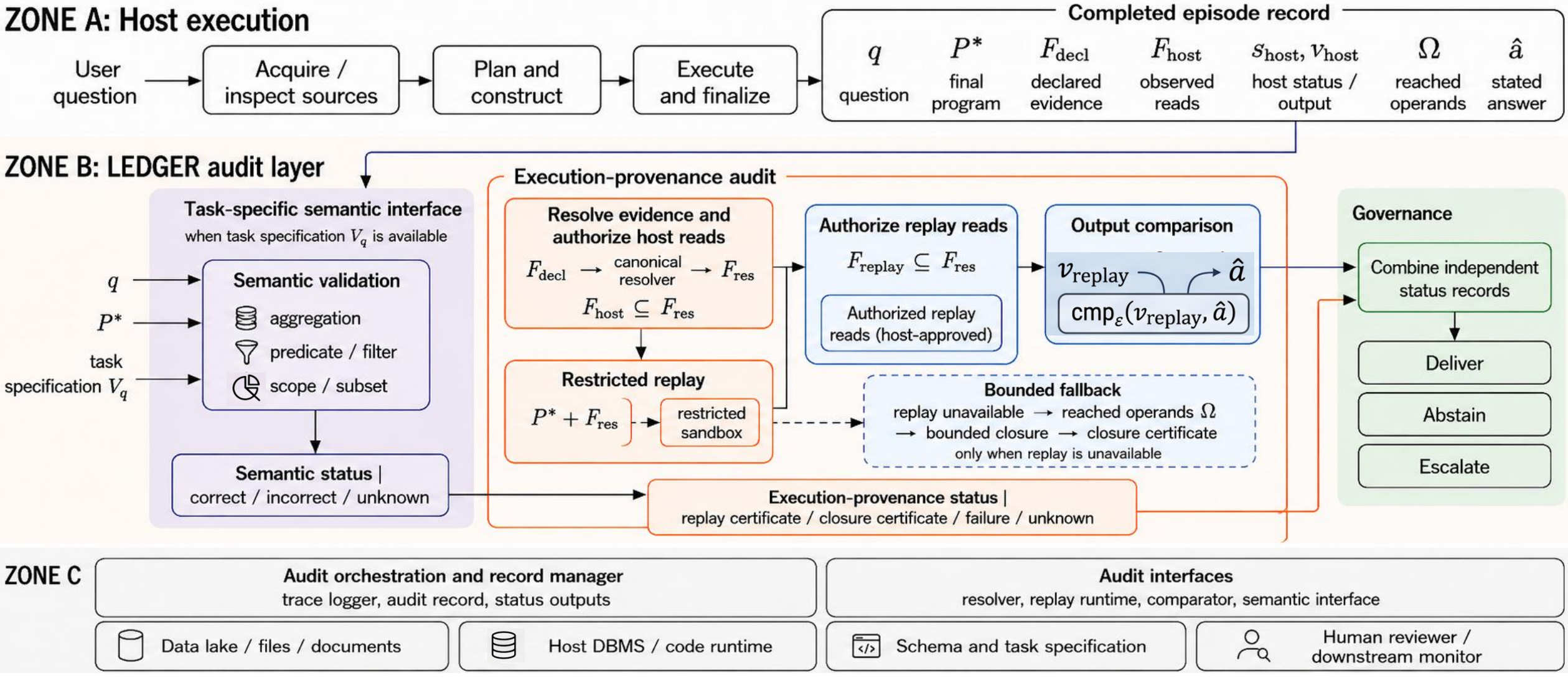

ZONE A: Host execution
User question
Acquire / inspect sources
Plan and construct
Execute and finalize
Completed episode record
q question
P^* final program
F_{decl} declared evidence
F_{host} observed reads
$s_{\text{host}}, v_{\text{host}}$ host status / output
Ω reached operands
$\hat{a}$ stated answer
ZONE B: LEDGER audit layer
Task-specific semantic interface
when task specification V_q is available
q
P^*
task specification V_q
Semantic validation
aggregation
predicate / filter
scope / subset
Semantic status | correct / incorrect / unknown
Execution-provenance audit
Resolve evidence and authorize host reads
F_{decl} → canonical resolver → F_{res}
$F_{\text{host}} \subseteq F_{\text{res}}$
Restricted replay
$P^* + F_{\text{res}}$
restricted sandbox
Authorize replay reads
$F_{\text{replay}} \subseteq F_{\text{res}}$
Authorized replay reads (host-approved)
Output comparison
v_{replay}
$\hat{a}$
$\text{cmp}_\varepsilon(v_{\text{replay}}, \hat{a})$
Bounded fallback
replay unavailable → reached operands Ω → bounded closure → closure certificate
only when replay is unavailable
Execution-provenance status | replay certificate / closure certificate / failure / unknown
Governance
Combine independent status records
Deliver
Abstain
Escalate
ZONE C
Audit orchestration and record manager
trace logger, audit record, status outputs
Audit interfaces
resolver, replay runtime, comparator, semantic interface
Data lake / files / documents
Host DBMS / code runtime
Schema and task specification
Human reviewer / downstream monitor